

MODULE 03

Software Testing Life Cycle (STLC) with AI

Assignment: Test Plan & Test Checklist Online Shopping Website

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Batch: 21st

Module: 03

Topic: Software Testing Life Cycle (STLC) with AI

Q1. Test Plan for an Online Shopping Website

1.1 Test Plan Overview

Project: Online Shopping Website

Testing Type: Functional, UI, Usability, Compatibility, Security & Performance

Testing Level: System Testing

Testing Approach: Manual Testing

Testing Environment: Web Browser

Application Type: E-commerce Website

Target Users: Registered and Guest Users

Test Priority: Critical features will be tested first

Test Data: Valid, Invalid, Boundary and Negative data

Defect Tracking: Bug/Issue Tracking System

Test Deliverables: Test Plan, Test Checklist, Test Cases, Defect Reports & Test Summary

1.2 Test Objectives

1. Verify that users can register successfully.
2. Verify that users can log in with valid credentials.
3. Verify that invalid login credentials are handled correctly.
4. Verify that users can browse and search products.
5. Verify that users can add and remove products from the cart.
6. Verify that product quantity and price calculations are correct.
7. Verify that users can complete checkout successfully.
8. Verify that payment processing works correctly.
9. Verify that order confirmation is generated after successful payment.
10. Identify and report defects before the website is released.

1.3 Scope of Testing

In Scope: User Registration; User Login/Logout; Product Search; Product Details; Product Categories; Add to Cart; Update Cart; Remove from Cart; Wishlist; Checkout; Address Management; Payment; Order Confirmation; Order History; UI Validation; Browser Compatibility

Out of Scope: Third-party payment gateway internal functionality; Server hardware testing; Physical delivery process; Internal bank transaction processing; Production database administration

1.4 Testing Types

Functional Testing:	Verify features work according to requirements
UI Testing:	Verify layout, buttons, fields and navigation
Usability Testing:	Verify the website is easy to use
Compatibility Testing:	Verify website works across browsers/devices
Security Testing:	Verify user data and authentication are protected
Performance Testing:	Verify acceptable response time under expected load
Regression Testing:	Verify existing functionality after changes
Negative Testing:	Verify system handles invalid inputs correctly
Boundary Testing:	Verify minimum/maximum input limits

1.5 Test Environment

OS: Windows 10/11
Browser: Google Chrome, Mozilla Firefox, Microsoft Edge
Device: Desktop/Laptop
Testing Method: Manual Testing

1.6 Entry Criteria

- 1. Requirements are finalized.
- 2. Test environment is available.
- 3. Application build is deployed.
- 4. Test data is available.
- 5. Major features are accessible.
- 6. Required test accounts are created.

1.7 Exit Criteria

- 1. All planned test cases are executed.
- 2. Critical and High severity defects are resolved or accepted.
- 3. Regression testing is completed.
- 4. Major functionality works as expected.
- 5. Test results are documented.
- 6. Test summary report is prepared.

1.8 Risks

Risk	Impact	Mitigation
Application downtime	High	Coordinate with development team
Unstable test environment	High	Maintain a stable QA environment
Payment gateway unavailable	High	Use test/sandbox payment gateway
Requirement changes	Medium	Update test cases regularly
Insufficient test data	Medium	Prepare data before testing
Browser compatibility issues	Medium	Test on multiple browsers

1.9 Deliverables

- 1. Test Plan
- 2. Test Checklist
- 3. Test Cases
- 4. Test Data
- 5. Defect Reports
- 6. Regression Test Results
- 7. Test Summary Report

Q2. Test Checklist

The following checklist contains 20+ validation points for each feature.

A. Login Checklist

SL	Test Area	Validation Point	Expected Result
1	Username	Enter valid registered email/username	User should be able to enter the value
2	Password	Enter valid password	Password should be accepted
3	Login	Enter valid username and password	User should log in successfully
4	Username Validation	Enter invalid username/email	Appropriate validation/error message should appear
5	Password Validation	Enter incorrect password	Appropriate error message should appear
6	Empty Fields	Leave username empty	Required field message should appear
7	Empty Fields	Leave password empty	Required field message should appear
8	Empty Fields	Leave both fields empty	Required validation messages should appear
9	Password Visibility	Click password visibility icon	Password should be shown/hidden correctly
10	Remember Me	Select Remember Me	Login session should behave according to requirement
11	Forgot Password	Click Forgot Password	Password recovery page should open
12	Account Lock	Enter incorrect password repeatedly	Account protection mechanism should work according to requirement
13	Special Characters	Enter special characters in username	Input should be handled safely
14	SQL Injection	Enter SQL injection-like input	Application should reject/safely handle malicious input
15	XSS	Enter script-like input	Script should not execute
16	Case Sensitivity	Test username/email with different letter cases	Behavior should follow requirements
17	Password Length	Enter password below minimum length	Validation should be displayed if applicable
18	Password Length	Enter password above maximum limit	System should handle input according to requirement
19	Login Button	Click Login with valid data	Login request should be processed
20	Session	Refresh page after login	User session should remain according to requirement
21	Logout	Click Logout after login	User should be logged out successfully
22	Navigation	Access protected page after logout	User should be redirected to login if authentication is required

B. Registration Checklist

SL	Test Area	Validation Point	Expected Result
1	Name	Enter valid full name	Name should be accepted
2	Email	Enter valid email address	Email should be accepted
3	Email Validation	Enter invalid email format	Validation message should appear
4	Password	Enter valid password	Password should be accepted
5	Confirm Password	Enter matching password	Confirmation should be accepted
6	Password Match	Enter different confirmation password	Password mismatch message should appear
7	Required Fields	Leave name empty	Required field message should appear
8	Required Fields	Leave email empty	Required field message should appear
9	Required Fields	Leave password empty	Required field message should appear
10	Required Fields	Leave all fields empty	Required validation messages should appear
11	Duplicate Email	Register using existing email	Duplicate account error should appear
12	Password Length	Enter password below minimum length	Appropriate validation should appear
13	Password Complexity	Enter weak password	System should show validation if complexity is required
14	Password Visibility	Click password visibility icon	Password should show/hide correctly
15	Phone Number	Enter valid phone number	Phone number should be accepted
16	Phone Validation	Enter invalid phone number	Validation message should appear
17	Special Characters	Enter unexpected special characters	System should validate/safely handle input
18	Terms & Conditions	Leave Terms checkbox unchecked	Registration should not proceed if acceptance is mandatory
19	Terms & Conditions	Accept Terms & Conditions	User should be allowed to continue
20	CAPTCHA	Enter incorrect CAPTCHA if implemented	Registration should be blocked
21	CAPTCHA	Enter correct CAPTCHA if implemented	Registration should continue
22	Submit	Submit valid registration data	Account should be created successfully
23	Verification	Verify email/OTP if implemented	Account should be verified successfully
24	Registration Message	Complete successful registration	Appropriate success message should appear
25	Login	Login using newly created account	User should be able to log in successfully

C. Checkout Checklist

#	Test Area	Validation Point	Expected Result
1	Cart	Proceed to checkout with valid product	Checkout page should open
2	Product	Verify selected product	Correct product should appear
3	Quantity	Verify product quantity	Correct quantity should be displayed
4	Price	Verify product price	Correct price should be displayed
5	Subtotal	Verify subtotal calculation	Subtotal should be calculated correctly
6	Discount	Apply valid discount/coupon	Discount should be applied correctly
7	Invalid Coupon	Apply invalid coupon	Appropriate error message should appear
8	Expired Coupon	Apply expired coupon	Coupon should be rejected
9	Shipping	Verify shipping charge	Correct shipping charge should be displayed
10	Total	Verify final order total	Total should be mathematically correct
11	Customer Name	Enter valid customer name	Name should be accepted
12	Address	Enter valid delivery address	Address should be accepted
13	Required Address	Leave required address field empty	Validation message should appear
14	Phone	Enter valid phone number	Phone number should be accepted
15	Invalid Phone	Enter invalid phone number	Validation message should appear
16	Payment	Select available payment method	Payment method should be selected
17	Payment Details	Enter valid payment/test details	Details should be accepted
18	Invalid Payment	Enter invalid payment details	Appropriate error should appear
19	Payment Failure	Simulate failed payment	Order should not be confirmed incorrectly
20	Payment Success	Complete successful payment	Order should be placed successfully
21	Order Confirmation	Verify confirmation message	Order confirmation should be displayed
22	Order ID	Verify order number/ID	Unique order ID should be generated
23	Email/SMS	Verify order notification if implemented	Confirmation notification should be sent
24	Stock	Purchase available product	Order should be processed successfully
25	Out of Stock	Attempt to purchase unavailable product	System should prevent unavailable purchase
26	Cart Update	Change quantity during checkout if allowed	Price and total should update correctly
27	Back Navigation	Go back during checkout	Cart/order information should remain consistent
28	Security	Verify sensitive payment information	Sensitive information should not be exposed unnecessarily
29	Refresh	Refresh checkout page	Application should handle refresh without corrupting order state
30	Order History	Check order after successful purchase	New order should appear in order history